

OPTIMISATION RESULTS

SEP and MTOjects Toy Objects

Four-fold cross-validation, final parameter selection, comparative interpretation and conclusions

Calibration sample: 40 low-foreground galaxies; four folds of 30 training and 10 held out

Optimisation: 40 trials per fold: 8 startup trials followed by 32 TPE-guided trials

Application target: 182-galaxy sample with six injected toy objects per galaxy

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Executive summary

Overall conclusion

SEP is the preferred primary method for the present Toy Objects workflow. Its cross-validated score was materially higher and its toy recovery was stronger. The revised MTOjects optimisation is nevertheless a genuine recovery from the earlier zero-mask failure: it now detects and removes toy objects, but does so with much lower pixel precision and substantial collateral masking. MTOjects should therefore remain a secondary, review-controlled method until its false-positive behaviour is reduced.

- SEP winner: fold 4; common-40 score 0.5416. Mean held-out score across four folds was 0.5026 (standard deviation 0.0880).
- MTOjects recovery winner: fold 3; common-40 score 0.2325, mean toy recall 50.7%, toy detection rate 53.3%, and mean masked fraction 9.07%.
- Only the MTOjects fold-3 candidate met every cross-fold feasibility gate. This supports its selection, but also shows that MTOjects parameter performance is less robust across folds than SEP.
- The very low MTOjects pixel precision (0.31%) is not a contradiction of toy detection: toy pixels occupy a very small part of each image, while the selected mask covers many non-toy pixels. The method recovers toys but with high collateral masking.

1. Experimental design

The same set of 40 galaxies with few foreground objects was used for both methods. In each of four rotations, 30 galaxies were used for parameter optimisation and the remaining 10 were held out for validation. Each fold used 40 Optuna trials: eight startup trials for broad exploration and 32 trials guided by the Tree-structured Parzen Estimator. Final fold candidates were also evaluated on a common 40-galaxy injection set to support a like-for-like selection.

The objective rewards recovery of injected toy objects while penalising false-positive masking, loss of usable galaxy data and excessive total mask coverage. A hard maximum masked-fraction limit of 15%

was applied in the MTOBJECTS recovery follow-up. The revised MTOBJECTS objective also explicitly makes non-detection infeasible, preventing the optimiser from preferring an empty mask merely because it avoids false-positive costs.

2. SEP optimisation results

Fold	Held-out score	Common-40 score	Maximum masked
1	0.5177	0.4991	13.6%
2	0.5809	0.5289	14.0%
3	0.3553	0.4862	12.3%
4 (selected)	0.5566	0.5416	12.3%

The selected fold-4 candidate did not have the single highest held-out score, but it had the best common-40 score. This is the appropriate selection criterion because every candidate was compared against the same 40-galaxy evaluation set. Fold 3 was notably weaker on its held-out group, but the remaining three held-out scores were close, suggesting generally useful transfer with one more difficult partition.

Selected SEP parameters

Parameter	Selected value	Parameter	Selected value
Detection image	Residual	Detection threshold	1.8577
Minimum area	5 pixels	Deblend thresholds	32
Deblend contrast	0.00179745	Background size	48
Filter size	1	Dilation radius	4 pixels
Maximum area	4,309 pixels	Maximum elongation	26.6013
Central exclusion	8 pixels	Winning fold	4

SEP interpretation

SEP provides the better balance of recovery and restraint. Its common-40 score is more than twice the MTOBJECTS score under their respective recovery objectives. The result supports SEP as the default automated foreground-mask generator, subject to visual review of outliers and galaxies with complex intrinsic structure.

3. MTOBJECTS recovery optimisation results

Fold	Held-out score	Common-40 score	Feasible across folds
1	0.1837	0.1958	No
2	0.2617	0.2070	No
3 (selected)	0.2442	0.2325	Yes
4	0.2135	0.2294	No

Fold 2 achieved the highest individual held-out score, while fold 4 approached the selected candidate on the common-40 score. Fold 3 was selected because it was the only candidate to satisfy the minimum toy-detection and toy-recall gates in every fold while remaining below the 15% maximum masked-fraction limit. This selection favours robustness over a single favourable split.

Selected MTOBJECTS parameters

Parameter	Selected value	Parameter	Selected value
Detection image	Original	Move factor	0.80516
Minimum distance	0.21516	Gaussian FWHM	0.42985
Background variance	0.00082103	Minimum area	5 pixels
Dilation radius	3 pixels	Maximum area	2,987 pixels
Maximum elongation	12.3811	Central exclusion	8 pixels
Alpha	1×10^{-6}	Winning fold	3

Common-40 performance of the selected candidate

Metric	Result	Interpretation
Optimisation score	0.2325	Best feasible common-set candidate
Mean toy recall	50.7%	About half of injected toy pixels recovered
Toy detection rate	53.3%	Just over half of toys detected
Mean pixel recall	45.5%	Moderate recovery of toy truth pixels
Mean pixel precision	0.31%	Most masked pixels are outside toy truth
Mean masked fraction	9.07%	Material image area removed
Maximum masked fraction	14.63%	Close to the 15% constraint
False-positive fraction	9.05%	Collateral mask dominates total mask

4. Comparative assessment

Criterion	SEP	MTOjects	Assessment
Common-40 score	0.5416	0.2325	SEP clearly stronger
Cross-fold robustness	Three strong held-out folds; one weaker	Only one candidate passed all gates	SEP more stable
Recovery behaviour	Strong recovery/restraint balance	Moderate recovery	SEP preferred
Collateral masking	Controlled below evaluated limits	False-positive fraction about 9%	MTOjects needs reduction
Recommended role	Primary automated method	Secondary diagnostic/review method	Do not combine blindly

Scores should be compared as decision evidence rather than as universal physical quantities: each score is a composite of recovery and penalty terms. The direction and magnitude nevertheless support the same operational conclusion because SEP combines a higher objective score with a more favourable recovery/collateral balance.

5. Conclusions

- The four-fold design is materially more defensible than four independent ten-galaxy optimisations because every galaxy contributes to validation while each parameter set is trained on 75% of the calibration sample.
- SEP has produced a credible, cross-validated parameter set for application to the full sample and should be treated as the baseline method.
- The MTOjects recovery follow-up solved the objective-function pathology: an empty mask is no longer considered a satisfactory result when toys are present.
- MTOjects now demonstrates genuine toy recovery, but its low precision and high false-positive area show that recovery is purchased at substantial cost to unaffected pixels.
- Neither optimisation should be interpreted as proof of perfect foreground removal on real contaminants; injected toys approximate foreground sources but cannot span all morphologies, surface-brightness contrasts and galaxy structures.

6. Recommended next steps

- Adopt SEP as the primary production mask and review the 40 filenames marked '_clean' first, because these are the calibration galaxies and therefore the most important check for overfitting or visually implausible masks.
- Retain MTOBjects outputs for comparison, targeted difficult cases and method-agreement analysis, rather than replacing SEP across all galaxies.
- For a further MTOBjects run, increase the penalty on non-toy masked pixels or impose a stricter mean-mask constraint, while preserving the explicit non-detection penalty and minimum recovery gates.
- Report both toy-level detection and pixel-level precision/recall. Toy detection alone can look satisfactory even when the mask includes a large amount of unrelated image area.
- Before final scientific use, visually classify failures and compare the original versus processed isophotes and bar-major profiles to ensure foreground masking has not altered the galaxy structure being measured.

7. Current application status

Status at document preparation

The optimisation results above are complete. The earlier SEP 182-galaxy application completed successfully. The newly aligned eight-panel MTOBjects batch created 181 of 182 reports; NGC3185 failed because no valid toy-injection candidates were available inside its investigated galaxy area. The visible launcher therefore stopped before starting the corresponding aligned SEP batch. Consequently, conclusions about the optimisation are final, but the newest aligned 182-galaxy comparative PNG set is not yet complete.

This application failure is a toy-placement edge case, not evidence that the selected MTOBjects parameters failed on NGC3185. It should be handled separately by allowing a controlled injection fallback or explicitly producing a no-toy diagnostic for that galaxy, followed by resumption of both batch stages.

8. Result provenance

MTOBjects final selection: mtojects toy recovery
followup\20260816_063455\mtobjects_toy_cross_validation_best.json. SEP final parameters were reconstructed from the preserved optimisation workbook and fold results after the original run directory was removed. Detailed numerical records are retained in documentation\Foreground Masking Optimisation Results.xlsx.

Metric values are rounded for readability. Full-precision values remain in the JSON, CSV and workbook artefacts.